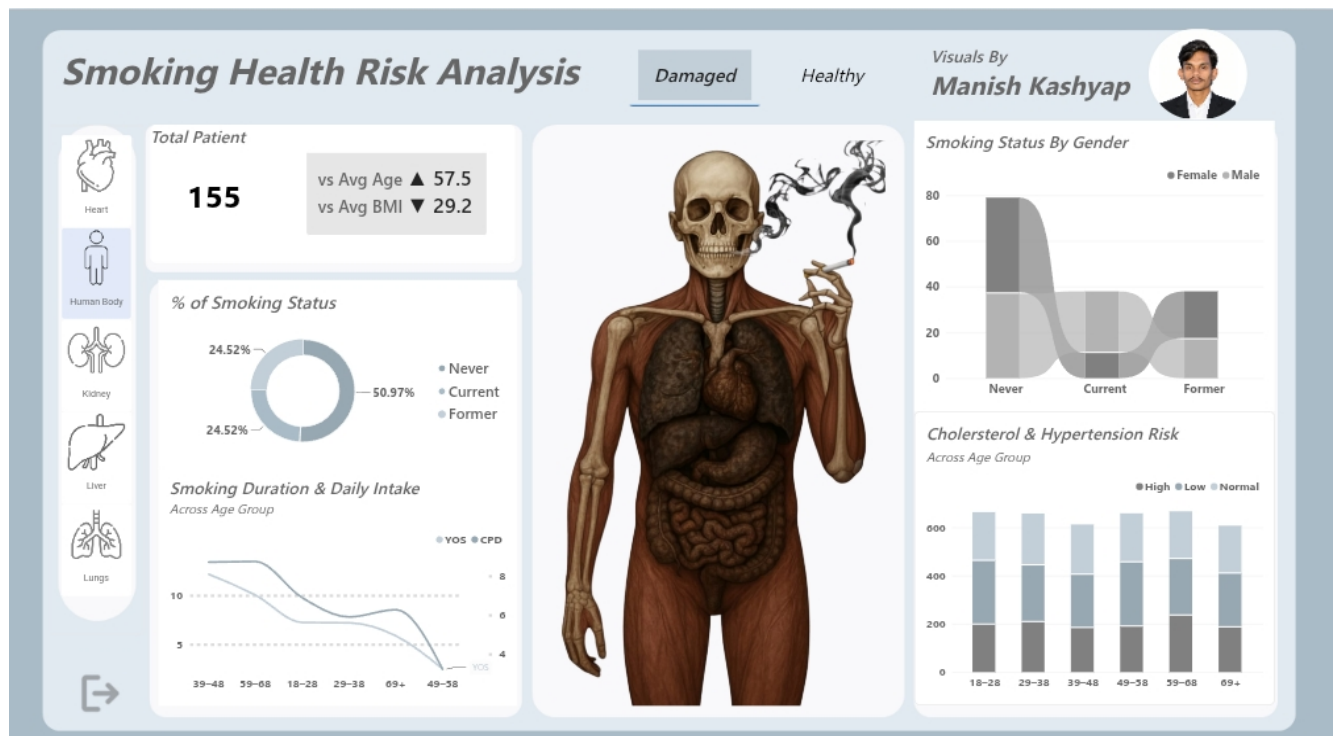


SMOKING HEALTH RISK ANALYSIS

Interactive Power BI Dashboard & Data Analytics Project Report



Prepared from the supplied project package

Dataset: 2,500 patient records • Power BI report: Smoking Health care.pbix • Dashboard default view: Human Body + Damaged

Analyst: Manish Kashyap

Purpose: Explore smoking status, smoking exposure, organ condition and related health-risk indicators through an interactive analytical dashboard.

1. Executive Summary

The supplied dataset contains **2,500 patient records** and 13 analytical fields covering demographics, smoking exposure, organ selection, organ condition, BMI, blood-pressure risk, cholesterol, family-history risk and alcohol consumption. Across the complete dataset, **897 records (35.9%)** are labelled Damaged and **1,603 (64.1%)** Healthy. The average age is **54.0**, average BMI is **29.6**, average cigarettes/day is **6.8**, and average years of smoking is **9.0**.

The Power BI report is designed around a condition slicer and an organ slicer. The saved/default dashboard state applies **Organ Condition = Damaged** and **Organ = Human Body**; in that filtered context the headline card shows **155 patients**, with an average age of **57.5** and average BMI of **29.2**. This distinction is important: dashboard card values are filter-dependent and should not be interpreted as whole-dataset totals.

Key findings from the supplied data

- Smoking status: Never smokers are the largest group (1,136; 45.4%), followed by Current (733; 29.3%) and Former (631; 25.2%).
- Organ condition: 35.9% of all records are labelled Damaged. The damage rate varies only modestly by organ, ranging from 32.7% to 37.7%.
- Age groups: the 18–28 group has the highest observed damage rate (40.0%), while 29–38 has the lowest (32.4%). These are descriptive differences, not proof of causation.
- Smoking exposure: Current smokers average 16.8 cigarettes/day and 22.2 years of smoking in this dataset.
- Blood-pressure and cholesterol indicators: High BP-risk records account for 629 patients, while 802 records have cholesterol at or above 240 in the supplied numeric scale.

Important interpretation note: this package does not document a clinical study protocol, sampling method or external source for the dataset. Therefore, the results should be treated as descriptive analysis of the supplied project data rather than medical evidence or population-level estimates.

2. Dataset Overview & Data Dictionary

The primary analytical table is **health_dataset.csv**. Supporting files provide the condition dimension, organ icons and organ/condition image URLs. The Power BI model also contains calculated logic for age grouping and comparison-to-overall-average cards.

Item	Value
Rows / patient records	2,500
Columns in primary dataset	13
Unique patients	2,500
Age range	18–89
Gender categories	Male, Female
Smoking categories	Never, Current, Former
Organ categories	Heart, Liver, Lungs, Kidney, Human Body
Condition categories	Healthy, Damaged
Missing values	0 in every primary column

Primary field definitions

Field	Type	Description / analytical role
Patient_ID	Integer	Unique patient/record identifier.
Age	Integer	Patient age in years.
Gender	Category	Gender label used for demographic comparisons.
Smoking_Status	Category	Never, Current or Former smoker status.
Years_of_Smoking	Integer	Reported/recorded years of smoking exposure.
Cigarettes_Per_Day	Integer	Daily cigarette intake measure.
Organ	Category	Organ/body category being evaluated.
Organ_Condition	Category	Healthy or Damaged condition label.
BMI	Numeric	Body mass index value supplied in the dataset.
BP_Risk	Category	Blood-pressure risk category: Low, Normal or High.
Cholesterol_Level	Numeric	Cholesterol measurement used for descriptive segmentation.
Family_History_Risk	Category	Yes/No family-history risk indicator.
Alcohol_Consumption	Category	Low, Moderate or High consumption category.

3. Data Quality, Preparation & Methodology

Data quality checks

Check	Result	Implication
Missing values	0 across 13 primary fields	No null-value treatment was required.
Patient ID uniqueness	2,500 unique IDs	Each row can be treated as one unique patient record in this project.
Age range	18–89	All records fall inside the supplied adult-age range.
Categorical completeness	All expected categories present	Suitable for slicers and grouped analysis.
Supporting condition table	2 rows: Healthy/Damaged	Provides the dashboard condition slicer values.
Supporting organ table	5 organ categories + icons	Supports organ navigation/slicer visuals.

Power BI / analytical logic

The supplied formula file defines an **Age Group** calculated column using six bands: 18–28, 29–38, 39–48, 49–58, 59–68 and 69+. Two measures, **vs Avg Age** and **vs Avg BMI**, calculate the current filtered average and compare it with the overall dataset average. An up/down indicator is then displayed on the dashboard card.

The dashboard contains a card, condition slicer, organ image slicer, donut chart, line chart, ribbon chart, column chart and supporting images. The supplied PBIX definition shows the condition slicer defaulted to **Damaged** and the organ slicer defaulted to **Heart** in the report definition; the supplied dashboard preview visually corresponds to a Human Body + Damaged view and displays 155 patients. This illustrates why saved filter state and preview state should be checked when presenting a final dashboard.

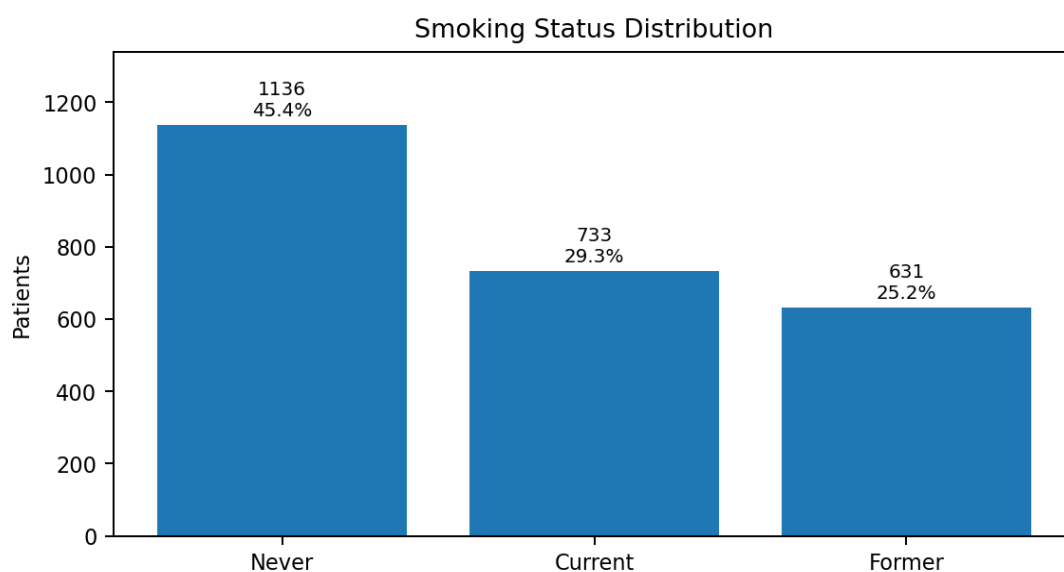
Analytical approach

1. Profile the dataset and validate completeness.
2. Establish overall KPIs before applying filters.
3. Compare smoking groups and smoking exposure measures.
4. Examine organ-level condition differences.
5. Segment results by age group, BP-risk category and cholesterol bands.
6. Reconcile the Power BI dashboard's filter-dependent KPIs with the underlying CSV.
7. Separate descriptive association from causal or clinical interpretation.

4. KPI Overview

KPI	Overall value	Interpretation
Total patients	2,500	Complete primary dataset.
Damaged records	897 (35.9%)	Records labelled Damaged.
Healthy records	1,603 (64.1%)	Records labelled Healthy.
Average age	54.0 years	Overall mean age.
Average BMI	29.6	Overall mean BMI.
Average cigarettes/day	6.8	Overall mean daily intake.
Average smoking duration	9.0 years	Overall mean years of smoking.
Average cholesterol	207.8	Overall mean supplied cholesterol value.

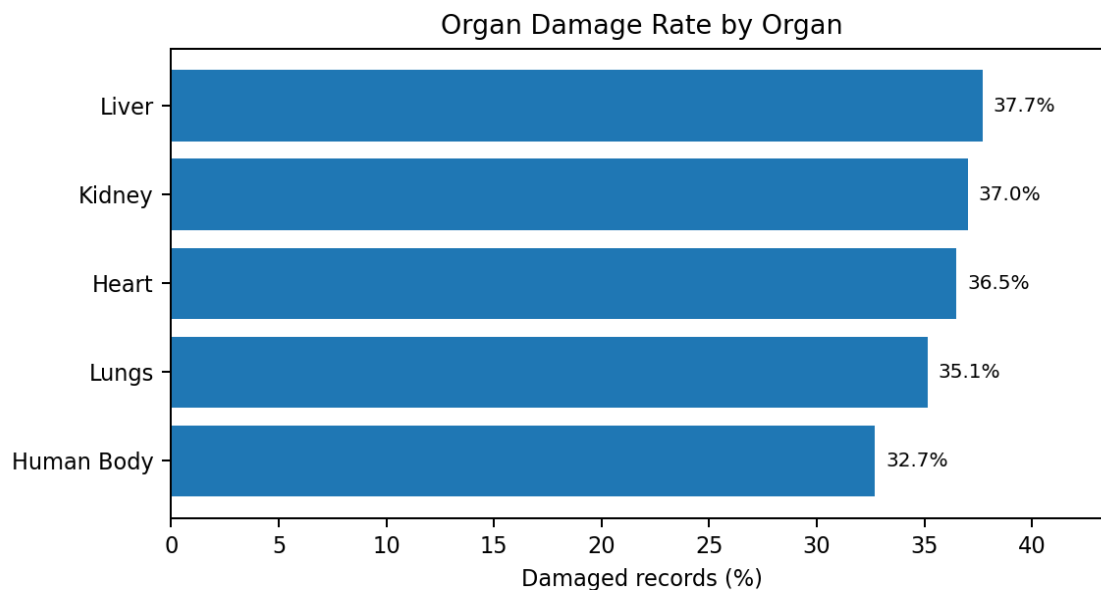
Smoking status distribution



Never smokers comprise 45.4% of records, while current smokers comprise 29.3%. The project therefore contains substantial representation of both exposed and non-exposed groups, but the dataset alone does not establish comparable real-world prevalence.

5. Organ Condition Analysis

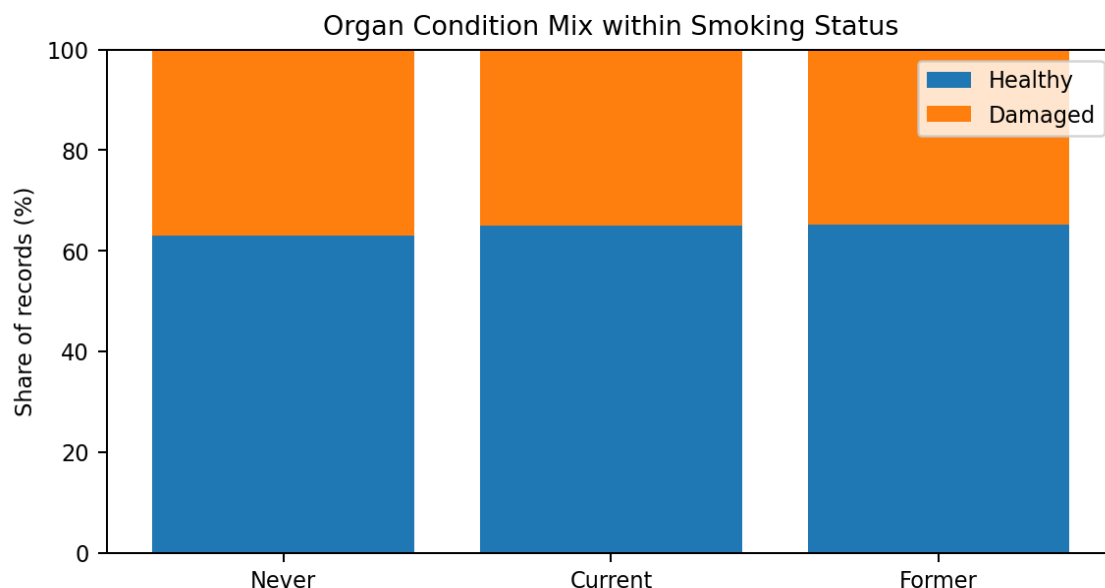
Across all organs, 897 of 2,500 records are labelled Damaged. Organ-level damage rates are relatively close to one another: the highest is Liver at 37.7%, while the lowest is Human Body at 32.7%.



Organ	Patients	Damaged	Damage rate
Heart	529	193	36.5%
Liver	549	207	37.7%
Lungs	478	168	35.1%
Kidney	470	174	37.0%
Human Body	474	155	32.7%

The relatively narrow spread in organ damage rates means that organ selection is more useful in the dashboard for focused exploration than for claiming a large difference between organs.

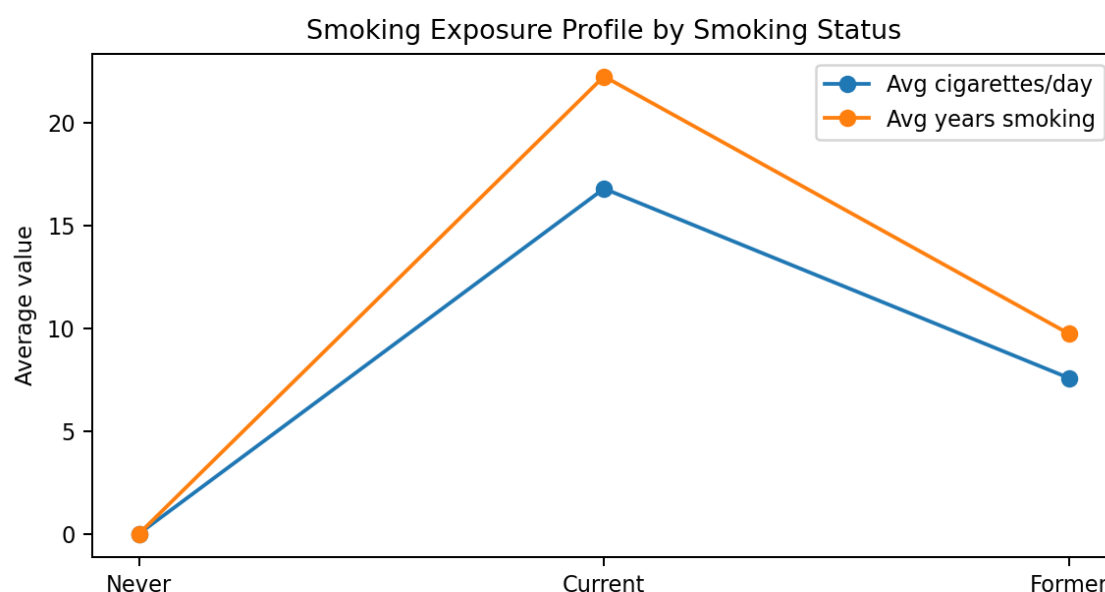
6. Smoking Exposure & Condition Analysis



Status	Patients	Avg years	Avg cigarettes/day	Avg BMI	Damage rate
Never	1,136	0.0	0.0	29.5	37.1%
Current	733	22.2	16.8	29.4	34.9%
Former	631	9.7	7.6	30.0	34.9%

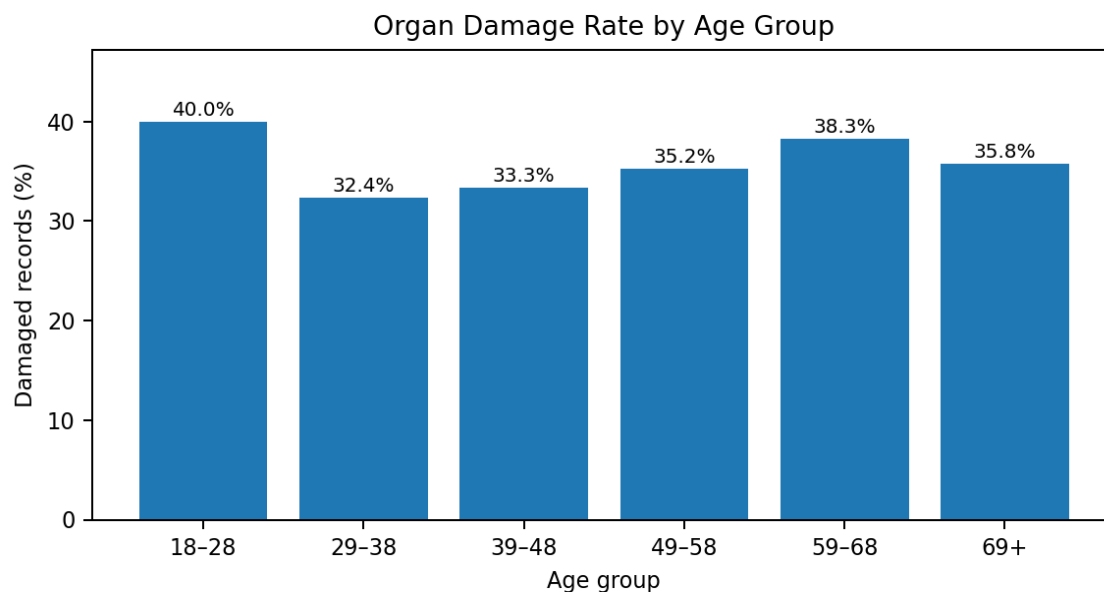
In the supplied records, current smokers have the highest average daily intake (16.8 cigarettes/day) and the longest average smoking duration (22.2 years). However, the observed damaged-record share is very similar across the three smoking-status groups (37.1% Never, 34.9% Current, 34.9% Former). This dataset therefore does not show a large descriptive separation in the binary organ-condition label by smoking status.

Because age, organ category, BMI, BP-risk, cholesterol and other variables may differ across groups, a causal interpretation would require a defined study design and statistical modelling. The dashboard is best used here for exploration and hypothesis generation.



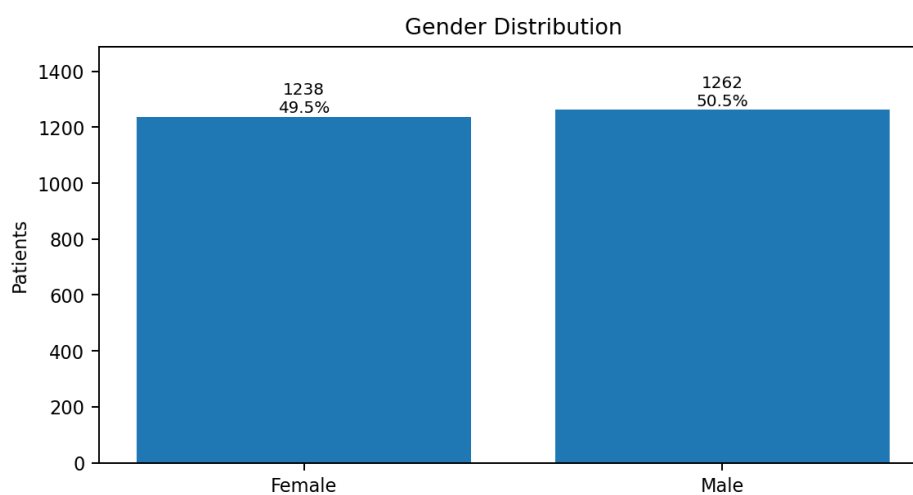
7. Demographic & Risk-Factor Analysis

Age-group analysis



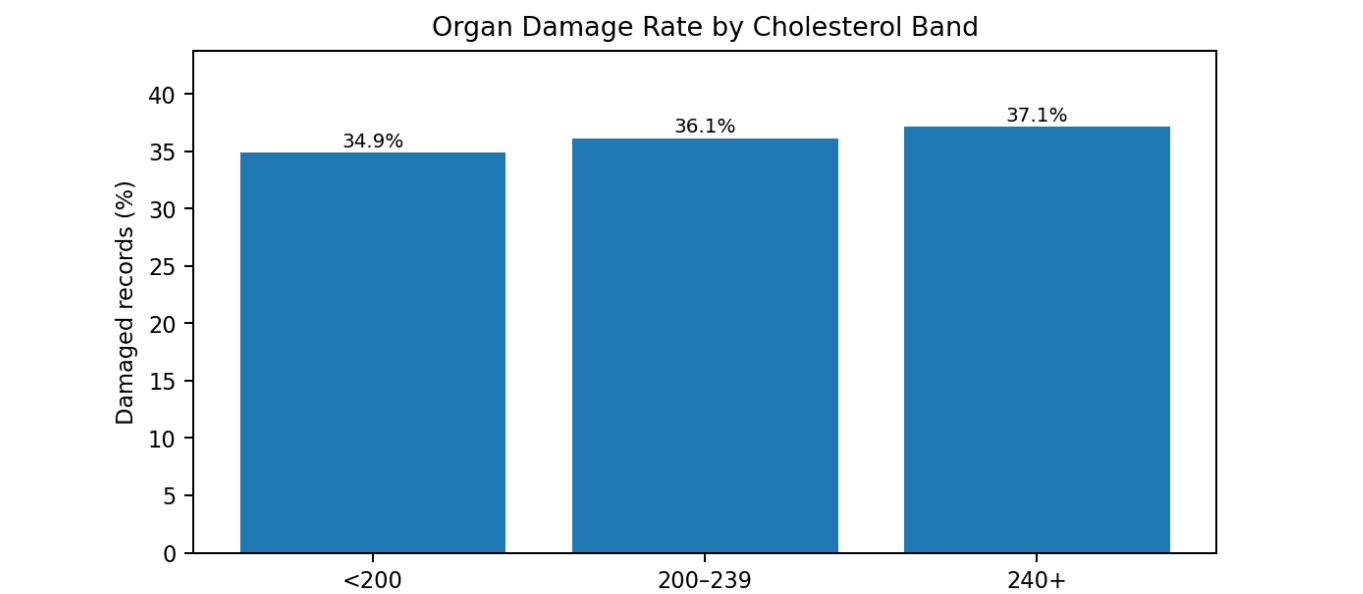
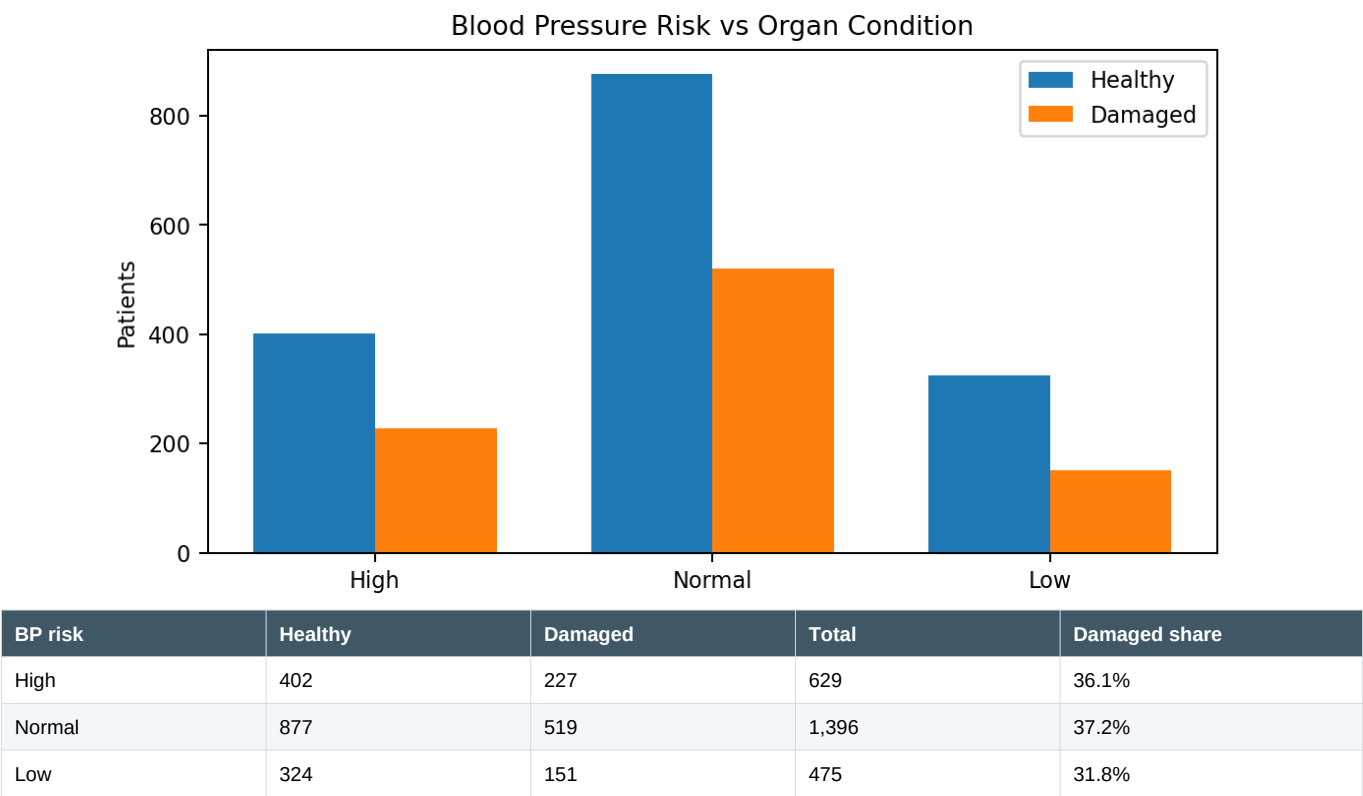
Age group	Patients	Damaged	Damage rate	Avg CPD	Avg BMI	Avg chol.
18-28	370	148	40.0%	7.2	29.4	208.0
29-38	343	111	32.4%	6.5	29.6	211.6
39-48	339	113	33.3%	7.1	30.2	202.8
49-58	349	123	35.2%	7.0	29.8	203.8
59-68	342	131	38.3%	7.3	29.1	206.5
69+	757	271	35.8%	6.4	29.7	210.6

Gender distribution



The dataset is close to balanced by gender: 1,262 Male (50.5%) and 1,238 Female (49.5%).

8. Blood Pressure & Cholesterol Analysis

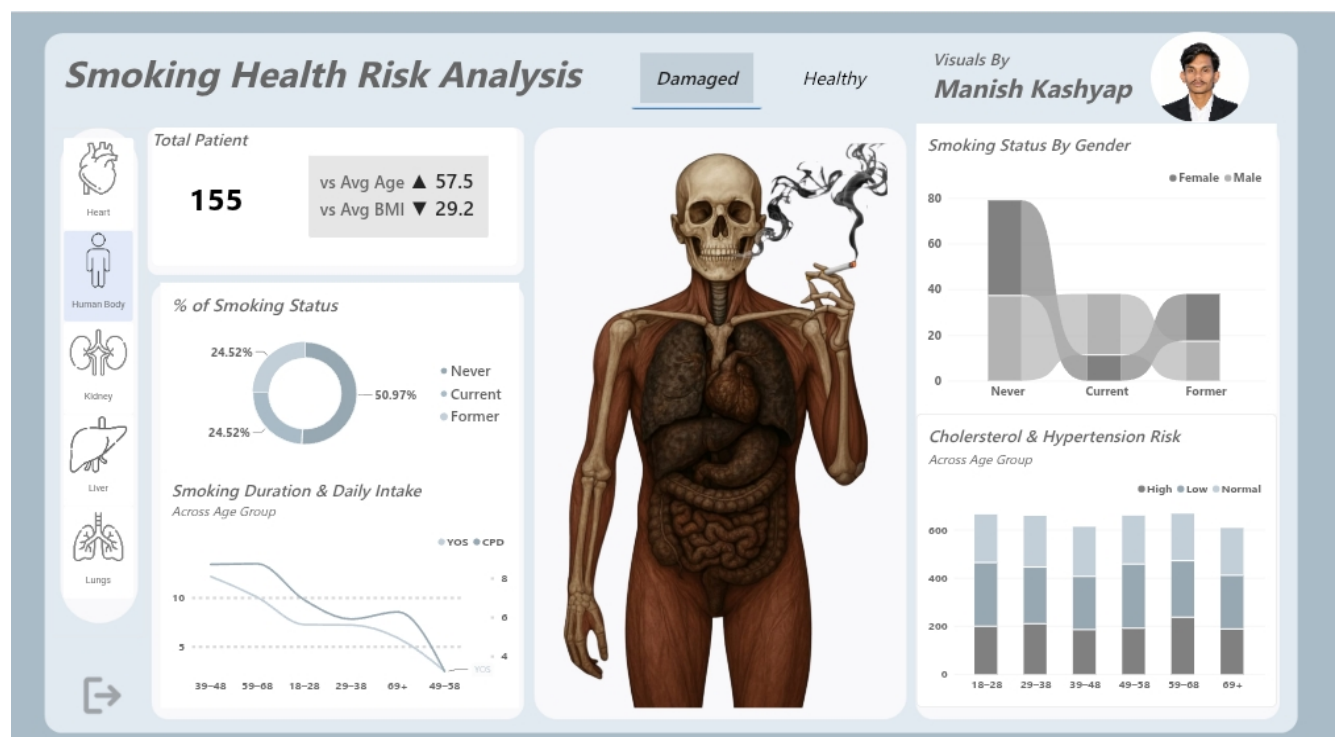


Using simple cholesterol bands, the observed damage share rises from 34.9% for values below 200 to 37.1% for values of 240 or above. This is a descriptive pattern in the supplied dataset; it should not be interpreted as a clinical risk estimate without additional context.

9. Power BI Dashboard & User Experience

The dashboard is built as a single-page interactive report with a healthcare-themed visual layout. Its main interaction model is filter-first: users can switch the organ condition and organ selection, then observe the KPI card and supporting visuals update.

Dashboard element	Purpose
Condition slicer	Switch between Healthy and Damaged records.
Organ slicer	Select Heart, Human Body, Kidney, Liver or Lungs using icon-based navigation.
KPI card	Shows patient count plus filtered average-age and average-BMI comparison measures.
Smoking-status donut	Shows the percentage distribution of Never, Current and Former status in the active filter context.
Smoking exposure line chart	Compares years of smoking and cigarettes/day across age groups.
Smoking-status ribbon chart	Shows the relationship between smoking status and gender counts.
Cholesterol/BP column chart	Displays average cholesterol across age groups, segmented by BP-risk category.
Central organ image	Provides visual context for the selected organ/condition.



Default preview interpretation: the supplied preview shows **Damaged** condition and a **Human Body** selection, giving a filtered patient count of 155. Within this context, the average age is 57.5 and average BMI is 29.2.

10. Default Dashboard Filter Context: Human Body + Damaged

This section reconciles the dashboard preview with the underlying dataset. When Organ = Human Body and Organ Condition = Damaged, there are **155 records**. This equals the complete count of Human Body records labelled Damaged.

Metric	Filtered value
Patients	155
Average age	57.5
Average BMI	29.2
Average cigarettes/day	6.5
Average years of smoking	7.1
Average cholesterol	207.2
High BP-risk share	21.3%
Family-history Yes share	29.0%

Smoking status within this filtered view

Smoking status	Patients	Share
Never	79	51.0%
Current	38	24.5%
Former	38	24.5%

This filtered section should be read as a dashboard drill-down, not as a separate sample. The records are a subset of the same 2,500-row dataset.

11. Key Findings & Analytical Implications

Finding	Evidence from supplied data	Analytical implication
Dataset is complete	0 missing values across primary fields.	Descriptive aggregations can be calculated without imputation.
Damaged records are a substantial subset	897 / 2,500 (35.9%).	Condition is suitable as a primary dashboard segmentation field.
Smoking exposure differs strongly by status	Current smokers average 16.8 cigarettes/day and 22.2 years exposure.	Exposure variables are useful for segmentation, even though the binary condition is not.
Organ damage rates are relatively close	Approx. 32.7%–37.7% across organs.	Avoid interpreting organ category alone as a large differentiator.
Age groups show descriptive variation	Damage rate ranges from 32.4% to 40.0%.	Age segmentation is useful for exploration; differences need statistical testing.
Risk indicators can be layered	BP risk and cholesterol are available alongside condition.	A future model can test multivariable associations rather than one-variable comparisons.

Recommended next analytical steps

- Add descriptive confidence intervals and significance tests for key group comparisons.
- Use multivariable logistic regression with Organ_Condition as the outcome and age, smoking status/exposure, BMI, BP risk, cholesterol, family history and alcohol category as predictors, if the dataset is intended for statistical modelling.
- Check class balance within each organ and demographic subgroup before interpreting percentages.
- Add tooltips that clearly state the active filter context so users do not confuse filtered KPIs with overall totals.
- Add a reset/clear-filters control and a small 'Current filters' panel to improve dashboard transparency.
- Document the dataset source, collection method, definitions and units, especially for cholesterol and the Healthy/Damaged label.

12. Limitations & Responsible Interpretation

- The dataset source, sampling strategy and collection period are not documented in the supplied project package.
- The Healthy/Damaged field is a binary project label; its clinical definition is not provided.
- No causal inference should be made from these descriptive cross-tabulations.
- The data may be synthetic, simulated or otherwise project-generated; it should not be assumed to represent the general population.
- The analysis does not adjust for confounding variables, selection effects or measurement error.
- Cholesterol and BP-risk values/categories are used exactly as supplied and are not interpreted as medical diagnoses.
- Dashboard values depend on slicer selections; filtered values must be distinguished from whole-dataset KPIs.

Responsible-use statement: This report is a data-analytics project document. It is not a medical report and should not be used to diagnose disease, estimate an individual's medical risk, or replace advice from a qualified healthcare professional.

Appendix A. Project Files Reviewed

File	Role in report
health_dataset.csv	Primary 2,500-row analytical dataset.
condition.csv	Healthy/Damaged condition dimension used by the Power BI slicer.
Organs.csv	Organ labels and icon URLs for the organ selector.
Image Dataset.csv	Organ + condition image URL mapping.
Measures and Columns Formula.txt	DAX logic for Age Group, vs Avg Age and vs Avg BMI.
Smoking Health care.pbix	Power BI report definition, visuals, slicers and model package.
Dashboard preview.jpeg	Visual reference for the supplied dashboard state.
Background Viz.png	Dashboard background asset.
power bi working.mp4	Project walkthrough/workflow recording supplied in the package.

Appendix B. Final project takeaway

The project demonstrates an end-to-end Power BI analytics workflow: structured health/smoking data → validation → calculated age groups and measures → interactive filtering → KPI cards → multi-dimensional visuals → descriptive insights. The strongest portfolio value is the combination of data preparation, DAX measures, interactive slicers and business-style storytelling, while the main improvement area is stronger documentation of data provenance and statistical methodology.

End of Report